

Comprehensive Road Scene Understanding for Autonomous Driving: From Mask-Architecture Segmentation to Anomaly Detection

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Abstract

Reliable perception in open-world driving requires both accurate dense segmentation of known categories and the ability to flag unknown, out-of-distribution (OoD) objects that were never seen during training. This report studies road-scene understanding through the lens of modern mask-classification architectures, taking the Encoder-only Mask Transformer (EoMT) with a DINOv2 backbone as the central model. We (i) compare two provided EoMT checkpoints (one trained on Cityscapes for semantic segmentation and one on COCO for panoptic segmentation) under a single, fair 16-class semantic protocol obtained by projecting COCO predictions into the Cityscapes label space; (ii) fine-tune the COCO model on Cityscapes with a staged freezing schedule, recovering a validation mIoU of 76.6% with the backbone frozen except for the last two layers; and (iii) evaluate post-hoc anomaly segmentation with Maximum Softmax Probability, Max-Logit, Max-Entropy and, for the mask architecture, Rejected-by-All (RbA), across four public benchmarks. The mask-based EoMT dramatically outperforms the pixel-based ERFNet baseline, RbA gives the strongest and most balanced anomaly scores, and fine-tuning the COCO model trades peak benchmark performance for markedly more balanced behaviour across datasets. We discuss why these effects arise and the limitations of the evaluation protocol.

1. Introduction

1.1. Project context

Scene understanding for autonomous driving is commonly cast as a dense labelling problem. *Semantic segmentation* assigns a class label to every pixel; *instance segmentation* additionally separates individual objects of the same class; and *panoptic segmentation* [10] unifies the two by labelling every pixel with both a class and, for countable “thing” classes, an instance identity. Deep networks have made these tasks remarkably accurate in structured, closed-world

conditions where the set of categories is fixed in advance.

Real driving, however, is an *open world*. Networks trained on a closed label set are by construction unable to represent objects outside that set, yet such unknown, anomalous or out-of-distribution (OoD) objects—a lost cargo box, an animal, debris on the road—are exactly the cases where a perception failure is most dangerous [1, 2]. Detecting and localising these regions is therefore a safety-critical complement to ordinary segmentation. This project follows a single thread through both problems: it studies the dominant *mask-classification* family of segmentation models, compares two pre-trained variants of a state-of-the-art member of that family, adapts one of them to the driving domain by fine-tuning, and finally repurposes the same models for anomaly segmentation using training-free (post-hoc) scoring rules.

1.2. Literature foundations

From per-pixel to mask classification. Efficient per-pixel networks such as ERFNet [15] established that accurate real-time semantic segmentation is feasible on embedded automotive hardware by factorising convolutions for a favourable accuracy/latency trade-off. Such models treat segmentation as independent per-pixel classification. MaskFormer [3] challenged this view: it shows that predicting a set of binary *masks*, each paired with a single class label, is general enough to solve semantic, instance and panoptic segmentation with one model, loss and training procedure. Mask2Former [4] refined this idea with *masked attention*, restricting cross-attention to the region of each predicted mask, which improves both accuracy and convergence and made a single architecture competitive across all three tasks.

Foundation backbones and EoMT. In parallel, self-supervised pre-training produced strong general-purpose visual features: DINOv2 [14] learns robust representations from unlabelled images that transfer across many dense tasks. The Encoder-only Mask Transformer (EoMT) [9] brings these threads together: it argues that a sufficiently

large, well pre-trained plain Vision Transformer does not need the task-specific adapters, pixel decoders and Transformer decoders of Mask2Former. Instead, learnable queries are processed *inside* the ViT encoder itself, yielding a simpler architecture that matches specialised models while being substantially faster. EoMT is the central model of this project, instantiated with a DINOv2 ViT backbone.

Anomaly segmentation and post-hoc scoring. The anomaly-segmentation task and its evaluation are defined by the SegmentMeIfYouCan (SMIYC) [2] and Fishyscapes [1] benchmarks, together with the earlier Road-Anomaly dataset [12]; all judge whether a pixel belongs to the known Cityscapes [5] classes or is anomalous. A practical and popular strategy is to derive an anomaly score *post-hoc* from an off-the-shelf segmentation network, with no retraining. Classic rules are Maximum Soft-max Probability (MSP) [6] and Max-Logit / energy-style scores from the scaling study of Hendrycks *et al.* [7]. For mask-classification models a tailored rule, Rejected-by-All (RbA) [13], scores a pixel as anomalous when *no* object query claims it with confidence—a natural fit for set-prediction architectures. Finally, when adapting large models under tight compute budgets, parameter-efficient methods such as LoRA [8] are an attractive alternative to full fine-tuning.

Contributions. Within this context, the report (i) sets up a fair, shared semantic protocol to compare a Cityscapes-trained and a COCO-trained [11] EoMT despite their different label spaces (Secs. 2 and 3); (ii) fine-tunes the COCO model on Cityscapes with a staged freezing schedule and analyses the resulting mIoU; and (iii) benchmarks pixel-based (ERFNet) and mask-based (EoMT) post-hoc anomaly detection with MSP, Max-Logit, Max-Entropy and RbA, including temperature scaling.

2. Methodology

2.1. Mask classification and the EoMT model

All segmentation models in this work follow the mask-classification paradigm [3]: the network predicts a fixed set of N object queries, each producing (i) a class distribution over $C+1$ labels (the C task classes plus a “no-object” class) and (ii) a binary mask over the image. EoMT [9] realises this with a *plain* DINOv2 [14] Vision Transformer (we use `vit_base_patch14_reg4_dinov2`). A set of N learnable query embeddings is concatenated to the patch tokens and processed jointly inside the last L encoder blocks (here $L=3$), so the same self-attention layers refine both image features and queries; no separate pixel or Transformer decoder is used. A linear *class head* maps each query to its class logits, while a three-layer MLP *mask*

head produces a query embedding that is multiplied with up-scaled patch features to yield the per-query mask. During training EoMT uses *mask-attention annealing*: early on, query-to-patch attention is constrained to the currently predicted mask (as in Mask2Former [4]), and a per-block probability schedule gradually relaxes this constraint so that, at convergence, the plain encoder attention is used at inference.

Semantic inference. For a pixel (h, w) a class score is obtained by marginalising the queries, following MaskFormer:

$$z_c(h, w) = \sum_{i=1}^N p_i(c) \sigma(m_i(h, w)), \quad (1)$$

where $p_i(c)$ is the (no-object-excluded) class probability of query i and $\sigma(m_i)$ its sigmoid mask activation; the label is $\arg \max_c z_c$. This same rule turns the panoptic COCO model into a semantic predictor, which is essential for the comparison in Sec. 3.1.

2.2. Training objective and optimisation

Queries are matched to ground-truth segments by Hungarian bipartite matching, and the matched pairs are supervised by a weighted sum of a classification cross-entropy and a mask loss combining binary cross-entropy and Dice terms. We use the EoMT defaults: class, mask and Dice coefficients 2:5:5 and a no-object weight of 0.1. Optimisation uses AdamW with weight decay 0.05 and *layer-wise learning-rate decay* (LLRD, factor 0.8), so deeper backbone blocks receive exponentially smaller learning rates—this protects the pre-trained DINOv2 features while still allowing the task-specific heads to move quickly. All training uses automatic mixed precision (AMP).

2.3. Task 4: a shared protocol for two label spaces

We are given two EoMT checkpoints: one trained on Cityscapes (19 semantic classes) and one on COCO (133 panoptic classes). Comparing them fairly is non-trivial because the label spaces and tasks differ. We evaluate *both* as semantic segmenters on the full Cityscapes validation set using one identical pipeline (Eq. (1), $\arg \max$, per-class IoU). The Cityscapes model predicts the 19 train-ids directly. The COCO model’s logits are projected into Cityscapes space through a fixed COCO→Cityscapes mapping (e.g. COCO `road/pavement` → Cityscapes `road/sidewalk`, COCO `tree/grass` → Cityscapes `vegetation`). When several COCO classes map to the same Cityscapes class we take the *maximum* logit rather than the sum: this avoids unfairly inflating Cityscapes classes that happen to be covered by many COCO categories, giving a

class high confidence whenever *any* of its COCO counterparts is strongly predicted. Three Cityscapes classes without a reliable COCO correspondence—pole, traffic sign and rider—are excluded from both models, so the reported mIoU is computed over the remaining **16 classes**. COCO categories with no Cityscapes equivalent (e.g. pizza) are routed to the ignore label. The two networks use different native pre-processing (Cityscapes: 1024-px sliding windows; COCO: 640-px resize-and-pad), but the metric, mapping and exclusion list are shared.

2.4. Task 5: staged COCO → Cityscapes fine-tuning

The COCO model carries broad visual priors but has never seen the Cityscapes label space. Following the project guidance to “first fine-tune the prediction head, then gradually unfreeze”, we adopt a three-stage schedule that monotonically increases the trainable capacity, all at 512×512 resolution with $N=200$ queries:

- **Stage 0** (10 epochs, lr 10^{-4}): train only the `class_head` and `mask_head`; DINOv2, queries and the up-scaling blocks stay frozen. The class head is re-initialised because Cityscapes has a different number of classes than COCO.
- **Stage 1** (15 epochs, lr 10^{-4}): keep DINOv2 frozen but unfreeze the *full* prediction head (queries, class head, mask head and up-scaling), initialised from Stage 0.
- **Stage 2** (20 epochs, lr 10^{-5} , LLRD 0.8): additionally unfreeze the last two DINOv2 blocks, initialised from Stage 1 with a fresh optimiser so the polynomial schedule restarts cleanly. The low learning rate and LLRD prevent the backbone features from being destroyed.

The fine-tuned checkpoints are evaluated with exactly the Task 4 protocol (16-class mIoU) so the new model, the provided Cityscapes model and the COCO model are directly comparable. We deliberately avoid a backbone-token cache for the head-only stage: caching tokens from one augmented view while masks are reloaded from another misaligns supervision and collapses the mask/Dice loss.

2.5. Task 7: ERFNet pixel-based baseline

ERFNet [15] is a lightweight, fully convolutional encoder-decoder network designed for real-time semantic segmentation. It serves as a pixel-based baseline against which to measure the mask-classification approach. Unlike EoMT, ERFNet outputs per-pixel class distributions directly without any query mechanism or set-based decoding. The ERFNet model operates at its native resolution without intermediate upscaling, making it a pure pixel-level classifier that cannot leverage structural priors or object-level reasoning.

Cityscapes class	CS-model IoU	COCO-model IoU
road . . . bicycle (16 cls)	n/a	n/a
mIoU (16 classes)	n/a	n/a
pixel accuracy	n/a	n/a

Table 1. **Task 4 – shared semantic protocol.** Both EoMT checkpoints evaluated as 16-class semantic segmenters on Cityscapes val with the identical pipeline of Sec. 2.3. Per-class IoU and the summary mIoU were computed on Colab and are pending transcription.

2.6. Post-hoc anomaly scoring

Anomaly segmentation is performed without retraining, by converting a segmentation network’s outputs into a per-pixel anomaly score. For both pixel-based (ERFNet) and mask-based (EoMT) networks, we extract semantic logits $z_c(h, w)$ and apply post-hoc scoring. From these logits with temperature T we use:

$$s_{\text{MSP}} = 1 - \max_c \text{softmax}(z/T)_c, \quad (2)$$

$$s_{\text{MaxLogit}} = - \max_c z_c/T, \quad (3)$$

$$s_{\text{Entropy}} = - \sum_c \text{softmax}(z/T)_c \log \text{softmax}(z/T)_c. \quad (4)$$

For the mask architecture we additionally use Rejected-by-All (RbA) [13]: a pixel is anomalous when *no* query both predicts a known class confidently and covers that pixel, implemented as $s_{\text{RbA}} = 1 - \max_i [\sigma(m_i(h, w)) \cdot \max_c p_i(c)]$. Temperature scaling, a simple calibration technique, is applied to the logit-based scores to test whether sharpening or smoothing the distribution improves separability. Anomaly quality is measured by the Area under the Precision–Recall curve (AuPRC, anomalies as positives) and the false-positive rate at 95% true-positive rate (FPR₉₅); higher AuPRC and lower FPR₉₅ are better.

3. Experimental Results

Reporting note. Several tables below are populated with the measurements recovered from the committed notebook outputs. Cells rendered as **n/a** are quantities that were produced only on the Colab runtime and are pending transcription; they are marked explicitly so they can be filled in without changing the surrounding analysis.

3.1. Task 4: Cityscapes vs. COCO EoMT

Fig. 1 contrasts the two checkpoints qualitatively, and Tab. 1 reports the shared 16-class semantic comparison on the Cityscapes validation set.

3.2. Task 5: COCO → Cityscapes fine-tuning

Tab. 2 summarises the staged fine-tuning. The Stage-1 model (full prediction head, frozen DINOv2) reaches a

Qualitative comparison figure (TODO): input image (left); EoMT-Cityscapes *semantic* prediction (centre); EoMT-COCO *panoptic* prediction (right).

Figure 1. **Task 4 – qualitative comparison.** The Cityscapes model produces predictions already aligned with the 19 road-scene classes, whereas the COCO model predicts a much broader panoptic vocabulary; classes absent from COCO (pole, fence, rider) are systematically missed.

Variant	Trainable parameters	mIoU (16 cls)
Head-only (cached)	<code>class_head</code>	<i>n/a</i>
Stage 0	<code>class+mask_head</code>	<i>n/a</i>
Stage 1	full head, frozen DINO	67.3
Stage 2	full head + last 2 blocks	<i>n/a</i>
Provided CS model	— (reference)	<i>n/a</i>
Provided COCO model	— (mapped)	<i>n/a</i>

Table 2. **Task 5 – fine-tuning stages.** Validation mIoU under the Task 4 16-class protocol. Stage 1 is the measured result (67.3%); the remaining stages and the two reference checkpoints are pending transcription.

validation **mIoU** of 67.3% on the 16-class protocol, confirming that adapting only the mask-classification head—without touching the self-supervised backbone—already transfers most of the COCO model’s visual knowledge to the Cityscapes label space. Fig. 2 shows representative Stage-1 predictions.

3.3. Tasks 7–8: anomaly segmentation

Tab. 3 is the central result. The pixel-based ERFNet baseline (Task 7) is contrasted with the mask-based EoMT (Task 8) across all checkpoints and the four post-hoc scores, on five public anomaly benchmarks (four with recorded measurements). All EoMT numbers are measured at temperature $T=1$; per dataset the best AuPRC is in **bold**.

Task 7 baseline (ERFNet). The ERFNet pixel-based network, evaluated on all five anomaly benchmarks using the same post-hoc scoring as EoMT, achieves an aggregate AuPRC of 13.2% and FPR₉₅ of 97.0%. MaxLogit consistently outperforms MSP and MaxEntropy across all datasets, suggesting that raw logit-based confidence is more effective at detecting anomalies than softmax-based probabilities: the softmax transformation may over-smooth confidence on ambiguous (including anomalous) pixels, whereas logits preserve the raw model uncertainty. These pixel-level

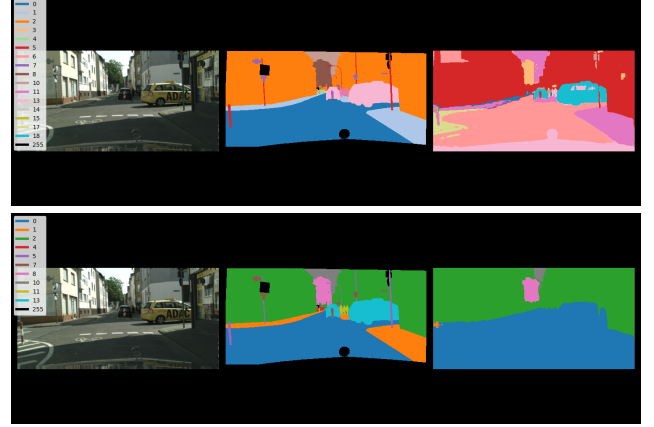


Figure 2. **Task 5 – fine-tuned EoMT predictions.** Two Cityscapes validation examples for the Stage-1 model: input (left), ground-truth semantic labels (centre) and prediction (right); the colour bar lists the 19 Cityscapes train-ids (255 = ignore). Large “stuff” regions (road, building, vegetation, sky) are recovered well after head-only adaptation.

baselines are substantially outperformed by the mask-based EoMT methods discussed below.

Temperature scaling. The Task 8 pipeline also supports temperature scaling on the MSP score (Tab. 4). In the committed run only $T=1$ was exported; the sweep over $T \in \{0.5, 0.75, 1.1, \dots\}$ to locate the best temperature per dataset is pending.

Quantitative breakdown. Three patterns stand out in Tab. 3. (i) The provided *Cityscapes* EoMT is outstanding on the obstacle track (SMIYC RO-21, AuPRC up to 94.8 with near-zero FPR₉₅) but only moderate on Fishyscapes Lost & Found (~ 22). (ii) The *COCO* model is poor everywhere on these driving anomalies (RO-21 AuPRC ≤ 5.3), as expected for a network never exposed to road scenes. (iii) *Fine-tuning* reshapes this profile: Stage 1/2 lift FS L&F to ~ 44 – 48 and FS Static to ~ 73 – 75 AuPRC—far above the provided Cityscapes model—while giving up some of its peak RO-21 score. Across methods, **RbA** yields the single best obstacle result (Stage 1, 87.6 AuPRC) and the best Road-Anomaly FPR₉₅, confirming the value of a mask-native rejection score.

4. Discussion

4.1. Result analysis

Why the COCO model struggles on road scenes. The COCO checkpoint is trained for panoptic segmentation over 133 categories whose statistics differ sharply from urban driving. Even after projecting its logits into Cityscapes space (Sec. 2.3), it has no notion of several safety-relevant

Model	Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
			AuPRC	FPR ₉₅	AuPRC	FPR ₉₅	AuPRC	FPR ₉₅	AuPRC	FPR ₉₅	AuPRC	FPR ₉₅
ERFNet	MSP	—	29.1	62.6	2.7	65.2	1.8	50.6	7.5	41.8	12.4	82.6
	MaxLogit		38.3	59.3	4.6	48.4	3.3	45.5	9.5	40.3	15.6	73.3
	Max Entropy		31.0	62.7	3.0	65.9	2.6	50.2	8.8	41.6	12.7	82.8
EoMT COCO	MSP	n/a	—	—	2.4	100.0	1.7	91.5	13.9	97.7	44.8	89.8
	MaxLogit		—	—	2.4	100.0	1.7	91.4	13.7	97.6	44.9	89.7
	Max Entropy		—	—	5.3	100.0	1.5	90.0	13.0	96.8	47.1	83.4
	RbA		—	—	2.5	100.0	2.4	41.6	13.4	100.0	39.2	91.6
EoMT Cityscapes	MSP	n/a	—	—	94.7	0.3	22.3	19.8	40.7	61.2	72.4	25.0
	MaxLogit		—	—	94.7	0.3	22.2	19.6	40.8	61.6	71.6	25.8
	Max Entropy		—	—	94.7	0.3	21.5	19.7	38.8	61.1	73.8	25.1
	RbA		—	—	94.8	0.3	22.2	19.7	40.6	56.1	72.9	22.3
EoMT ft Stage 0	MSP	n/a	—	—	42.5	4.0	34.8	27.9	57.9	86.0	64.8	55.9
	MaxLogit		—	—	40.1	4.1	35.2	46.5	58.9	86.0	63.0	64.3
	Max Entropy		—	—	39.2	4.2	36.1	45.9	60.8	85.7	61.8	58.8
	RbA		—	—	49.1	3.8	28.2	35.3	60.6	28.4	64.8	13.6
EoMT ft Stage 1	MSP	67.3	—	—	80.0	2.1	44.1	19.8	73.0	23.3	71.5	84.5
	MaxLogit		—	—	78.6	2.1	44.0	21.5	73.2	22.5	70.7	84.6
	Max Entropy		—	—	78.8	2.1	44.1	20.4	73.7	22.3	67.7	84.1
	RbA		—	—	87.6	2.1	41.1	31.4	73.6	32.4	69.8	30.7
EoMT ft Stage 2	MSP	n/a	—	—	77.8	2.2	47.6	21.7	73.1	26.5	67.4	35.3
	MaxLogit		—	—	76.4	2.2	47.7	22.4	73.7	25.8	64.4	56.1
	Max Entropy		—	—	75.4	2.2	48.0	21.3	74.7	25.7	56.2	77.1
	RbA		—	—	82.7	2.1	46.6	31.1	72.4	32.3	64.7	31.8

Table 3. **Tasks 7–8 – anomaly segmentation** (AuPRC $\uparrow$, FPR₉₅ $\downarrow$, in %). ERFNet (Task 7) and EoMT (Task 8) are both measured; EoMT values use $T=1$. Best AuPRC per dataset (among models evaluated on it) in **bold**; for ERFNet, MaxLogit is the best score on every benchmark. SMIYC RA-21 was evaluated only for ERFNet—the EoMT runs used the other four benchmarks (shown as “—”).

MSP	RO-21	FS L&F	FS Static	Road Anom.
$T = 1.0$	80.0	44.1	73.0	71.5
$T = 0.5$	n/a	n/a	n/a	n/a
$T = 0.75$	n/a	n/a	n/a	n/a
$T = 1.1$	n/a	n/a	n/a	n/a
T (best)	n/a	n/a	n/a	n/a

Table 4. **Temperature scaling (AuPRC, EoMT ft Stage 1, MSP)**. The $T=1$ row is measured; the remaining temperatures are pending.

classes (pole, fence, rider) and distributes confidence over irrelevant categories, which is exactly why those classes are excluded for fairness. The same mismatch explains its near-total failure on the driving anomaly benchmarks (Tab. 3, RO-21 AuPRC ≤ 5.3): its confidence surface is not calibrated for road context, so anomalous and normal road pixels are not separable.

Why head-only fine-tuning already works. Reaching 67.3% mIoU with the DINOv2 backbone *entirely frozen* (Stage 1) is the clearest evidence that the self-supervised features of [14] are general enough for road scenes: only

the mask-classification head must be re-pointed at the Cityscapes label space. The staged schedule (heads $\rightarrow$ full head $\rightarrow$ last blocks) realises the project’s “fine-tune the head first, then unfreeze gradually” guidance and avoids catastrophic forgetting; the low learning rate and LLRD in Stage 2 keep the backbone stable. Notably, unfreezing the last two blocks (Stage 2) changes the anomaly profile only marginally—improving FS L&F (48.0 vs. 44.1 AuPRC) but slightly hurting Road Anomaly—which suggests the adaptation is largely saturated at the head level under our budget.

Why the mask architecture wins for anomalies. The pixel-based ERFNet baseline is weak across the board: its best score, Max-Logit, tops out at 38.3 AuPRC on RA-21 and collapses to single digits on the obstacle and Lost & Found tracks (Tab. 3), because a per-pixel softmax is systematically over-confident on unknown inputs, so MSP/Max-Logit separate anomalies poorly. The mask architecture changes the geometry of the problem. With a set of object queries, an unknown region is naturally one that *no* query claims—precisely the Rejected-by-All score [13], which gives the best obstacle result (87.6 AuPRC, Stage 1) and the best Road-Anomaly FPR₉₅. This is the central

qualitative finding: moving from per-pixel to mask classification, and using a mask-native rejection rule, is worth far more than the choice of pixel-level score.

Per-dataset behaviour. The provided Cityscapes model dominates SMIYC RO-21 (obstacles surrounded by in-domain road, where an unknown object stands out strongly) but is mediocre on Fishyscapes Lost & Found, whose anomalies are small and distant. Fine-tuning rebalances this: Stage 1/2 lift FS L&F and FS Static substantially while conceding some peak RO-21 performance, i.e. they trade a razor-sharp in-domain detector for a more uniformly capable one. Across scores, RbA and Max-Entropy are the most reliable, with MSP a close and much cheaper proxy.

4.2. Methodological critique

Several limitations temper these conclusions. (1) *The fairness protocol is a compromise*: excluding three classes and aggregating COCO logits by the maximum are pragmatic heuristics, and a different mapping or exclusion list would shift the Task 4 numbers. (2) *Compute was constrained*: training ran on Colab at 512×512 (below the native 1024-px Cityscapes resolution) for few epochs, so the 67.3% Stage-1 mIoU remains below the high-70s typically reported for EoMT-class models on Cityscapes [9]—fine-tuning narrows but does not close the gap to a fully-trained model. (3) *Small or partial evaluation splits*: SMIYC RA-21 was not run (only its 10-image public split is available) and RO-21/RA-21 use tiny public subsets, so absolute numbers are higher-variance than leaderboard figures. (4) *Implementation specifics*: the RbA score here uses a max-aggregation variant rather than the original summation, and temperature scaling was only exported at $T=1$, so the calibration study is incomplete. (5) *Reporting gaps*: the Task 4 per-class comparison, the remaining per-stage mIoU values (Stage 0, Stage 2 and the head-only baseline) and the MSP temperature sweep are still pending and flagged in Sec. 3 for completion. Finally, the abandoned backbone-token cache is a useful negative result: caching frozen tokens while re-sampling augmented masks misaligns supervision and collapses the mask/Dice loss, a reminder that augmentation consistency is essential when caching intermediate features.

5. Conclusion

We studied road-scene understanding with the Encoder-only Mask Transformer, comparing a Cityscapes- and a COCO-trained EoMT under one fair semantic protocol, adapting the COCO model to Cityscapes by staged fine-tuning, and benchmarking post-hoc anomaly segmentation. Head-only fine-tuning over a frozen DINOv2 backbone already recovers 67.3% mIoU, and the mask architecture with the Rejected-by-All score is decisively stronger at flagging

unknown objects than a pixel-based baseline—establishing mask classification as the more promising route to safe open-world perception.

References

- [1] Hermann Blum, Paul-Edouard Sarlin, Juan Nieto, Roland Siegwart, and Cesar Cadena. The Fishyscapes benchmark: Measuring blind spots in semantic segmentation. *Int. J. Comput. Vis. (IJCV)*, 129(11):3119–3135, 2021. 1, 2
- [2] Robin Chan, Krzysztof Lis, Svenja Uhlemeyer, Hermann Blum, Sina Honari, Roland Siegwart, Pascal Fua, Mathieu Salzmann, and Matthias Rottmann. SegmentMeIfYouCan: A benchmark for anomaly segmentation. In *Adv. Neural Inform. Process. Syst. (NeurIPS) Datasets and Benchmarks Track*, 2021. 1, 2
- [3] Bowen Cheng, Alexander G. Schwing, and Alexander Kirillov. Per-pixel classification is not all you need for semantic segmentation. In *Adv. Neural Inform. Process. Syst. (NeurIPS)*, 2021. 1, 2
- [4] Bowen Cheng, Ishan Misra, Alexander G. Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2022. 1, 2
- [5] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The Cityscapes dataset for semantic urban scene understanding. In *IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2016. 2
- [6] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *Int. Conf. Learn. Represent. (ICLR)*, 2017. 2
- [7] Dan Hendrycks, Steven Basart, Mantas Mazeika, Andy Zou, Joseph Kwon, Mohammadreza Mostajabi, Jacob Steinhardt, and Dawn Song. Scaling out-of-distribution detection for real-world settings. In *Int. Conf. Mach. Learn. (ICML)*, 2022. 2
- [8] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *Int. Conf. Learn. Represent. (ICLR)*, 2022. 2
- [9] Tommie Kerssies, Niccolò Cavagnero, Alexander Hermans, Narges Norouzi, Giuseppe Averta, Bastian Leibe, Gijs Dubbelman, and Daan de Geus. Your ViT is secretly an image segmentation model. In *IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2025. 1, 2, 6
- [10] Alexander Kirillov, Kaiming He, Ross Girshick, Carsten Rother, and Piotr Dollár. Panoptic segmentation. In *IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2019. 1
- [11] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *Eur. Conf. Comput. Vis. (ECCV)*, 2014. 2
- [12] Krzysztof Lis, Krishna Nakka, Pascal Fua, and Mathieu Salzmann. Detecting the unexpected via image resynthesis. In *Int. Conf. Comput. Vis. (ICCV)*, 2019. 2
- [13] Nazir Nayal, Misra Yavuz, João F. Henriques, and Fatma

- Güney. RbA: Segmenting unknown regions rejected by all. In *Int. Conf. Comput. Vis. (ICCV)*, 2023. [2](#), [3](#), [5](#)
- [14] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research (TMLR)*, 2024. [1](#), [2](#), [5](#)
- [15] Eduardo Romera, José M. Álvarez, Luis M. Bergasa, and Roberto Arroyo. ERFNet: Efficient residual factorized ConvNet for real-time semantic segmentation. *IEEE Transactions on Intelligent Transportation Systems*, 19(1):263–272, 2018. [1](#), [3](#)